A Sparse-Sensing Spatial Digital Twin for Learning Environmental Impacts of Non-Networked Appliances in a Single Room

Yun-You Lin
Department of Computer Science and Information Engineering
National Changhua University of Education
Changhua, Taiwan
yunitrish0419@gmail.com

Chang-Pei Yi
Department of Computer Science and Information Engineering
National Changhua University of Education
Changhua, Taiwan

Hui-Yu Shen
Department of Electrical Engineering
Chienkuo Technology University
Changhua, Taiwan

Abstract—Practical room-scale digital twins must often operate with sparse sensors and non-networked appliances that expose no telemetry. This paper presents a single-room spatial digital twin that estimates temperature, humidity, and illuminance fields from eight corner sensors while modeling conventional air conditioners, manual windows, and ordinary lights. The proposed design combines an appliance-aware bulk-plus-local field model, photometric lamp falloff, lightweight one-bounce diffuse illuminance reflection, active-device power calibration, trilinear residual correction, least-squares appliance-impact learning, and an optional hybrid residual neural layer. Across eight canonical scenarios, the base estimator achieves average field MAE of 0.0474/0.1765/2.0269 for temperature, humidity, and illuminance, compared with 0.1723/0.4633/54.9052 for IDW. A leave-one-scenario-out hybrid residual evaluation further reduces MAE to 0.0017/0.0059/0.1407. On a preliminary seven-day bedroom snapshot dataset, calibration reduces held-out pillow-point MAE from 0.8967/4.1286/309.0142 to 0.1676/0.3939/16.6450. The evaluation separates controlled full-field evidence, public task-aligned comparisons, and real sparse-calibration checks. These results indicate that sparse corner sensing can support an interpretable and trainable indoor twin when physical structure, calibration, and residual learning are assigned complementary roles.

Index Terms—spatial digital twin, sparse sensing, indoor environment modeling, non-networked appliances, reduced-order model, sensor calibration

I. Introduction

Smart homes and smart buildings require indoor environmental awareness for comfort assessment, energy management, and device control. A practical system should estimate conditions at relevant regions such as a room center, window-side zone, or bed location, not only at installed sensors. This is difficult because dense sensing is intrusive, while ordinary rooms also contain conventional appliances that do not report state, power, or operating parameters.

This work targets a common but under-instrumented digital-twin setting: the room contains influential physical assets, but those assets are not necessarily connected to a building management system. The twin must therefore infer appliance state effects from sparse environmental observations, support what-if queries, and keep its claim boundary clear. Pure interpolation ignores appliance geometry and directionality, while a purely local field model misses whole-room settling after air-conditioner activation. We therefore use a reduced-order appliance-aware model as the primary estimator, sensor residuals for calibration, and neural learning only for structured residual correction rather than for an end-to-end black-box field predictor.

The contributions are: (i) a three-factor room digital twin with bulk-plus-local appliance influence functions, photometric lamp falloff, and one-bounce diffuse illuminance reflection; (ii) a sparse-sensor calibration pipeline combining active-device power fitting, trilinear residual correction, and before/after impact learning; (iii) a hybrid residual pathway that preserves the reduced-order estimator while refining systematic error; and (iv) an implementation and validation hierarchy covering synthetic full-field scenarios, ablations, leave-one-scenario-out residual learning, window-condition cases, public task-aligned datasets, and preliminary real-bedroom snapshots.

II. Related Work

Digital twins provide computational representations of physical systems and are increasingly used in manufacturing, infrastructure, and smart buildings [1], [2]. Building-oriented reviews show strong attention to BIM/BMS connectivity and system architecture, but sparse-sensor spatial estimation in ordinary rooms remains less explored [3]. For control-oriented indoor modeling, reduced-order

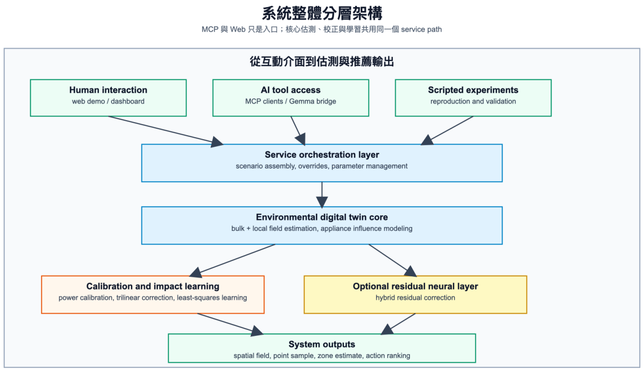

Fig. 1. Layered architecture of the proposed single-room spatial digital twin.

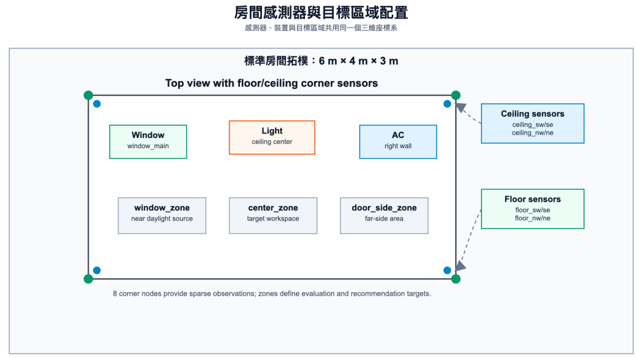

Fig. 2. Room topology, appliance placement, target zones, and the eight floor/ceiling corner sensors.

and grey-box approaches remain attractive because they are interpretable and tractable compared with CFD [4]–[6]. Zonal and hybrid indoor models preserve dominant spatial non-uniformity at lower cost [7]–[9], while data-assimilation studies show that finite observations can reconstruct indoor temperature and humidity fields and that sensor placement matters [10], [11]. The present work combines these strands by coupling sparse sensing with appliance-aware influence bases and residual calibration. IDW [12] is used as a geometry-free interpolation baseline. The service interface is secondary to the modeling contribution, but the same estimator is exposed through local APIs, a web demo, and an MCP-compatible tool layer [13].

III. Method

A. Room, Devices, and Field Model

The canonical room is a 6.0 m by 4.0 m by 3.0 m rectangular room with eight corner sensing nodes, four on the floor and four on the ceiling. Each node measures temperature, humidity, and illuminance. The room contains three evaluation zones and three appliance classes: an air conditioner, a window, and a light. The web demo represents these appliances as wall or ceiling markers and allows their activations to be toggled.

For metric $m \in \{T, H, L\}$ at point $\mathbf{p} = (x, y, z)$ and elapsed time t, the estimated field is

$$F_m(\mathbf{p}, t) = B_m^{\text{bulk}}(t) + B_m^{\text{local}}(\mathbf{p}, t) + \sum_j I_{j,m}(\mathbf{p}, t) + C_m(\mathbf{p}). \tag{1}$$

The bulk term captures whole-room settling, local terms preserve spatial gradients, $I_{j,m}$ represents appliance influence, and C_m is fitted from sparse sensor residuals. Each appliance influence is modulated by

$$E_j(\mathbf{p}, t) = a_j(1 - e^{-t/\tau_j}) D_j(\mathbf{p}) G_j(\mathbf{p}), \tag{2}$$

where a_j is activation, τ_j is a response time constant, D_j is distance attenuation, and G_j is directional gain. The air conditioner lowers temperature and humidity, the window uses outdoor temperature, humidity, and sunlight, and the light primarily raises illuminance with a small heat gain. The lamp direct term uses a cosine beam weight derived from beam angle, an inverse-square distance scale with bounded gain, and line-of-sight obstruction. Window daylight remains an envelope-based daylight term in the reported experiments; an aperture view-factor option is implemented but not used as the default because it increased window-family illuminance error in the current canonical setup.

Direct window and lamp terms underestimate indirect fill light near floors, walls, and furniture. Instead of full ray tracing, illuminance adds a lightweight one-bounce diffuse term

$$I^{\text{refl}}(\mathbf{p}, t) = \sum_{s \in \mathcal{S}} \rho_s \bar{I}_s A_s^{\text{rel}} e^{-\|\mathbf{p} - \mathbf{c}_s\| / \ell_s} \gamma_s(\mathbf{p}) V_s(\mathbf{p}), \tag{3}$$

where surfaces $\mathcal{S}$ include room boundaries and active furniture faces, ρ_s is reflectance, $\bar{I}_s$ is direct illuminance at surface center $\mathbf{c}_s$, A_s^{rel} is normalized area, ℓ_s is attenuation length, γ_s is diffuse alignment, and V_s is an obstruction term.

B. Sparse Calibration and Learning

Given predicted sensor value $\hat{y}_{i,m}$ and observed value $y_{i,m}$, residual $r_{i,m} = y_{i,m} - \hat{y}_{i,m}$ is used in two steps. First, active-device power scales are fitted by least squares over sensor residuals. Second, a trilinear correction is fitted:

$$\begin{aligned} C_m(x, y, z) = c_{0,m} + c_{1,m}X + c_{2,m}Y + c_{3,m}Z + c_{4,m}XY \\ + c_{5,m}XZ + c_{6,m}YZ + c_{7,m}XYZ, \end{aligned} \tag{4}$$

where X, Y, Z are normalized room coordinates. This cannot recover high-frequency turbulence, but it corrects bias, first-order gradients, and corner interactions from the eight observations. The trilinear form is deliberately matched to the sensing layout. With floor and ceiling sensors at all four horizontal corners, the eight observations constrain the eight coefficients without introducing a high-order surface that would overfit sparse points. The correction should therefore be interpreted as a low-frequency bias and gradient adjustment, not as a substitute for dense flow or lighting measurements. This design keeps the estimator identifiable from the available observations while still letting appliance geometry and dynamics remain in the explicit physical layer.

For non-networked devices, before/after sensor deltas supervise impact learning. The learning event is stored as an appliance-state record rather than as a free-text action label: it includes a record id, device name, applied device state, merged runtime device specifications, indoor baseline, outdoor boundary conditions, furniture state, elapsed time, and before observations. For an air conditioner, the applied state may include activation, mode, setpoint, fan speed or strength, outlet angles, and sweep settings. After observations from the same sensors complete the record. Given metric deltas $\Delta \mathbf{y}_m$ and appliance envelopes in matrix X, coefficients are estimated by

$$\boldsymbol{\beta}_m = \arg\min_{\boldsymbol{\beta}} \|\Delta \mathbf{y}_m - X\boldsymbol{\beta}\|_2^2. \tag{5}$$

For residual learning, point features include normalized coordinates, elapsed time, indoor/outdoor conditions, base estimates, device activations, calibrated powers, appliance envelopes, and air-conditioner operating state such as mode, setpoint, fan strength, outlet angles, and sweep settings. The hybrid residual layer predicts $R_m(\mathbf{p}, t; \theta_m)$ and returns $F_m^{\text{hybrid}} = F_m + R_m$. Temperature and humidity residual traces can be Fourier low-pass filtered before training because thermal and moisture responses are typically smoothed by heat capacity, air mixing, moisture exchange, and dehumidification. Illuminance uses raw residual labels because lighting, daylight, shadows, occlusion, and reflection can create physically meaningful short-timescale changes that low-pass filtering would remove.

The data flow in Fig. 3 is intentionally shared between training and runtime use. During learning, raw corner-sensor records, device events, outdoor boundary conditions, and scenario metadata are first time-aligned and normalized, then assembled into the same scenario state used by the service layer. This state is evaluated by the reduced-order estimator to produce base fields and residual labels. The pipeline then branches into least-squares impact learning from before/after sensor deltas and optional hybrid residual training from point-level residual samples, producing learned coefficients, validation summaries, and a checkpoint. At runtime, a user, web interface, or MCP client supplies the current baseline, outdoor conditions, device and furniture states, elapsed or steady-state time, and a query point or target zone. The service rebuilds the scenario state, evaluates the nominal fields, applies sparse correction and the optional residual checkpoint, and returns point or zone estimates. Action ranking is defined only after a decision sample scope is specified, either a single query point or a cluster/zone sample, and after a complete three-factor temperature,

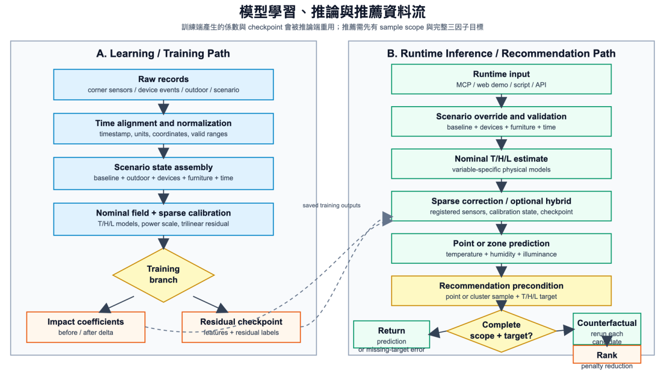

Fig. 3. Shared learning, inference, and recommendation data flow. Learned impact coefficients and the optional residual checkpoint are saved from the training path and reused by the runtime estimator.

humidity, and illuminance target/tolerance vector is supplied. Without those inputs, the service returns estimates only and does not emit recommendations. Given scope and target, each candidate device action is converted into a counterfactual scenario and passed through the same inference path; actions are sorted by predicted comfort-penalty reduction.

Candidate control actions are evaluated as model-based counterfactuals. Let $S = \{\mathbf{p}_k\}_{k=1}^K$ be the decision sample scope, where $K = 1$ is a point query and $K > 1$ is a cluster or target-zone sample. For metric m, the scope value is $q_m(S) = K^{-1} \sum_k F_m(\mathbf{p}_k, t)$. A weighted comfort penalty is computed from normalized deviations beyond target tolerances for temperature, humidity, and illuminance, and each action is ranked by current penalty minus simulated post-action penalty. AC actions carry a fuller operating state rather than only an on/off flag: cooling, drying, heating, or fan mode; temperature setpoint; fan speed or numeric fan strength; horizontal and vertical outlet angles; and fixed or sweeping airflow. This is recommendation ranking, not closed-loop control; causal validation requires before/after intervention trials.

IV. Implementation and Experimental Setup

The prototype is implemented as a zero-dependency Python package with modules for entities, scenarios, field simulation, impact learning, hybrid residual training, IDW comparison, action ranking, service APIs, an MCP endpoint, and a rotatable web demo. The shared service layer still supports the reproduction scripts and browser demo, while the MCP endpoint is intentionally narrowed to five runtime tools: environment initialization, elapsed-time or steady-state point sampling, before/after impact-learning records, direct window input, and point-target action ranking from registered devices. The ranking endpoint requires the point coordinates plus a complete temperature, humidity, and illuminance target; missing target fields produce no recommendation. The initialization tool registers the runtime state used by subsequent calls, including the base scenario, indoor baseline, outdoor boundary conditions, device and furniture states, AC mode/setpoint/fan/airflow controls when present, default elapsed or steady-state timing, and the estimator choice.

The service layer is intentionally kept separate from the estimator. A command-line script, the MCP endpoint, and the browser demo call the same scenario and sampling functions, so evaluation results are not produced by a separate code path from the interactive prototype. The browser interface supports metric switching, appliance toggling, AC mode/setpoint/fan-speed/outlet-angle controls, indoor baseline editing, season-weather-time presets, outdoor boundary overrides, point sampling, target-zone summaries, impact-learning output, and estimator switching between the base and hybrid residual fields when a saved checkpoint is available. This separation also reduces an implementation risk common in demonstration-driven digital twins: the visual interface can change without changing the numerical estimator, and the estimator can be tested from scripts without relying on browser state. The MCP layer is consequently treated as a deployment interface for the same service functions rather than as a separate modeling contribution. In the present paper, this matters because reported MAE values remain tied to reproduction scripts, while MCP point samples, learning records, direct window inputs, and action rankings expose the same runtime estimator without turning validation utilities such as IDW comparison or the full window matrix into MCP-facing claims.

All synthetic full-field experiments use the canonical room and $16 \times 12 \times 6$ grid, giving 1152 spatial samples per field. Scenarios are evaluated after an 18-minute settling interval with outdoor temperature 33.0°C, relative humidity 74.0%, and sunlight 32000 lux. The eight canonical scenarios are idle, AC only, window only, light only, AC+window, window+light, AC+light, and all active. Dense synthetic truth is generated by deterministic appliance power adjustments and sensor observations are synthesized from truth sensor predictions with fixed index-based perturbations: temperature adds $0.08((i \bmod 4) - 1.5)$, humidity adds $0.3((i \bmod 4) - 1.5)$, and illuminance adds $3.0((i \bmod 4) - 1.5)$. The main reproduction scripts are run_demo.py, run_hybrid_residual_experiment.py, run_submission_readiness_experiments.py, run_bedroom_weekly_simulation.py, and run_public_dataset_model_comparison.py.

Because no single dataset covers room geometry, dense 3-D ground truth, appliance state, and real sparse observations at once, the evaluation uses the layered claim boundary in Table I. We label the validation items as E1–E9 so that each numerical claim is tied to one evidence layer: E1–E6 are controlled synthetic or robustness experiments, E7 is a real-bedroom sparse-calibration check, E8 is a protocol for future before/after recommendation validation, and E9 is the public task-aligned benchmark.

TABLE I
Validation Scope and Claim Boundary

Evidence item	Supports	Boundary
E1–E4 synthetic full-field, IDW, ablation, and impact-learning checks	3-D reconstruction, baseline comparison, and component contribution under controlled truth	Simulated room and deterministic perturbations
E5 48-case window matrix	Sensitivity to outdoor temperature, humidity, sunlight, and opening ratio	Boundary-condition study, not measured weather deployment
E6 hybrid residual splits	Residual learnability under default, no-Fourier, and leave-one-scenario-out settings	Scenario-family robustness, not arbitrary-room generalization
E7 seven-day bedroom snapshots	Sparse calibration improves a held-out pillow point	One reference point; no dense real 3-D ground truth
E8 recommendation intervention protocol	Defines measured improvement, success rate, and top-1 regret	Protocol only; not counted as completed validation
E9 SML2010 and CU-BEMS task-aligned benchmarks	External positioning on compatible time-series or zone-level prediction tasks	Not direct validation of the full spatial twin

This separation is important because good performance on one layer does not automatically validate the others.

V. Evaluation

A. E1–E4 Synthetic Reconstruction and Impact Checks

Across the eight canonical scenarios, the base estimator achieves average field MAE of 0.0474 for temperature, 0.1765 for humidity, and 2.0269 for illuminance. IDW gives 0.1723, 0.4633, and 54.9052, respectively. Table II summarizes the main comparisons and ablations. Removing appliance-aware feedback produces larger temperature and illuminance errors; removing one-bounce reflection or active-device calibration mainly hurts illuminance. The no-trilinear variant can be lower on dense synthetic humidity and illuminance because trilinear correction enforces sparse sensor consistency rather than guaranteed dense-field monotonic improvement. This last point is important for interpreting the ablation table. The trilinear correction is fitted from corner observations, so it is optimized for consistency with sparse sensors and for the real-bedroom calibration setting. It is not optimized directly against every dense grid point in the synthetic room. A lower dense-field MAE in the no-trilinear row therefore does not imply that sparse calibration is unnecessary; it indicates that sparse correction and dense synthetic truth are different objectives. For deployment, the estimator must

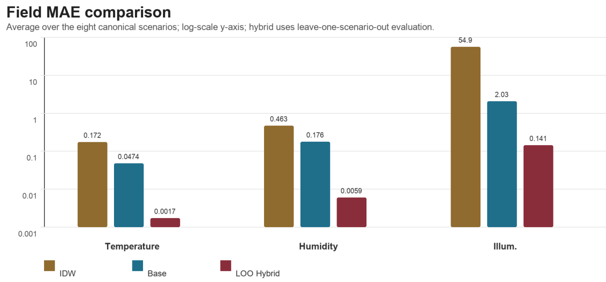

Fig. 4. Average field MAE for IDW, the base estimator, and leave-one-scenario-out hybrid residual correction. The y-axis is logarithmic.

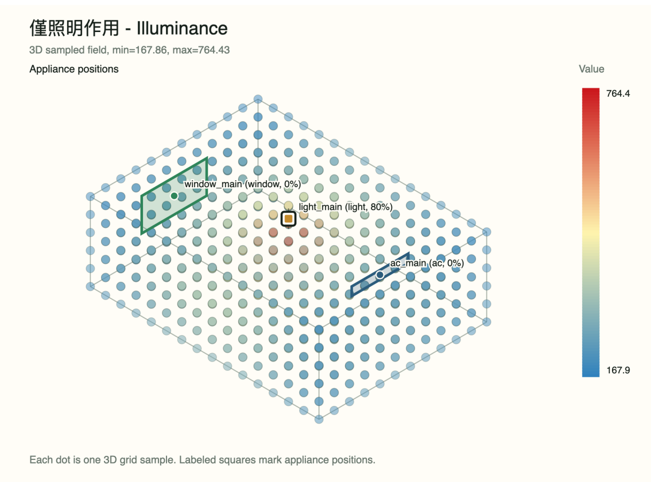

Fig. 5. Example 3-D illuminance field for the light-only scenario, showing localized lighting influence and reflected fill.

respect measured corner observations first, then use the appliance-aware model to extrapolate the field between them.

For impact learning, the AC-only scenario learns negative coefficients for temperature and humidity and near-zero illuminance effect. The light-only scenario learns a strongly positive illuminance coefficient, weak positive heat gain, and near-zero humidity effect. These directions match the intended physical interpretation of the appliances.

B. E6 Hybrid Residual Correction

The default held-out residual experiment trains on six scenarios and tests on light_only and all_active, yielding 576 training and 192 test samples. With Fourier denoising for temperature and humidity residual traces, hybrid MAE is 0.0020/0.0051/0.1370, compared with base MAE of 0.0474/0.1765/2.1757 on the same split. Illuminance is intentionally left unfiltered because its fast changes can correspond to source switching, daylight boundary changes, shadows, and reflections rather than noise. Disabling Fourier gives 0.0021/0.0057/0.1370, showing that the illuminance improvement is not caused by frequency-domain smoothing. In leave-one-scenario-out evaluation, each fold trains on seven scenarios and tests on one scenario, with 672/96 train/test samples. The average LOO hybrid MAE is 0.0017/0.0059/0.1407, corresponding to approximately 96.41%, 96.66%, and 93.06% reductions relative to the base estimator.

TABLE II
Synthetic Field MAE and Robustness Summary

Setting	Temp.	Hum.	Illum.
IDW baseline	0.1723	0.4633	54.9052
Raw nominal	0.1312	0.0842	3.5183
No reflection	0.0472	0.1762	2.4296
No calibration	0.0493	0.1772	3.3631
No trilinear	0.0446	0.0274	0.9849
Full base	0.0474	0.1765	2.0269
LOO hybrid avg.	0.0017	0.0059	0.1407

TABLE III
Real-Bedroom Snapshot Comparison

Case	Temp.	Hum.	Illum.	Penalty
Raw pillow MAE	0.8967	4.1286	309.0142	–
Calib. pillow MAE	0.1676	0.3939	16.6450	0.0911
Morning calib.	0.2372	0.3695	8.6247	0.2371
Afternoon calib.	0.1358	0.4890	11.2235	0.0812
Night calib.	0.0913	0.2193	46.6993	0.0459
Sleep calib.	0.2060	0.4979	0.0326	0.0000

C. E7 Real-Bedroom Snapshot Validation and E8 Intervention Protocol

We further evaluate calibration on a user-supplied bedroom dataset. The room is a 4.0 m by 4.6 m by 3.2 m bedroom with wall-mounted AC, southeast-facing window, ceiling light, desk lamp, bed, desk, and cabinets. The dataset covers 2026-04-14 to 2026-04-20, with 09:00, 15:00, 22:00, and 02:00 snapshots each day, for 28 cases. Each case provides eight corner observations, appliance activations, outdoor boundary information, and a pillow-location reference observation. Corner sensors are used for active-device calibration and trilinear correction; the pillow point is held out.

Table III shows that calibration reduces pillow-point MAE from 0.8967°C, 4.1286% relative humidity, and 309.0142 lux to 0.1676°C, 0.3939%, and 16.6450 lux. Sleep snapshots use a 0 lux illuminance target with 5 lux tolerance so dark sleep conditions are not penalized. These snapshots validate sparse-sensor calibration and point-level estimation, not the causal efficacy of recommendations; physical recommendation validation requires before/after intervention trials comparing measured comfort-penalty reduction against predicted improvement.

TABLE IV
Selected Task-Aligned Benchmark Results

Task	Metric	Pers.	LinReg	Proposed
S3 15min temp.	MAE	0.233	0.093	0.071
S3 15min temp.	Corr.	0.000	0.909	0.950
S3 60min temp.	MAE	0.563	0.212	0.190
S3 60min illum.	RMSE	21.93	22.76	19.34
S1 15min illum.	MAE	3.42	4.02	5.35
C2 15min illum.	MAE	1.36	1.79	7.70
S2 15min hum.	MAE	0.20	0.21	0.83

D. E5 Window Matrix and E9 Public Benchmarks

The window matrix contains 48 cases from four times of day, three weather states, and four seasons. It evaluates the window as an environmental variable rather than a binary switch, and the direct-input mode accepts outdoor temperature, outdoor humidity, sunlight, and opening ratio without discretizing the outside condition. Representative results include winter-rain-night (T_o = 11.0°C, H_o = 78.0%, sunlight 15.2 lux, window-zone illuminance 68.97 lux), spring-cloud-morning (21.5, 70.0, 5005.0, 93.22), and summer-sun-noon (37.0, 71.0, 36000.0, 243.71).

For external comparison, we evaluate task-aligned modes on SML2010 [14] and CU-BEMS [15]. Because these datasets do not provide the same room geometry or dense 3-D ground truth, tasks are restricted to compatible two-point time-series or zone-level response prediction. Compared with persistence and linear regression, the proposed model is strongest on compound tasks with appliance or external-condition transitions and on longer-horizon temperature prediction. Task-family analysis separates this behavior more clearly: SML2010 S3 is the main advantage because event-delta targets need change direction; S1 pure illuminance is a weakness because short-window light levels strongly favor persistence; S2 is mixed, with longer-horizon temperature benefits but humidity scale mismatch. CU-BEMS shows a different boundary: C1/C3 often beat linear regression, C2 lighting is weak, and no CU-BEMS task beats persistence because zone-level building data have strong temporal inertia. For example, SML2010 S3 15 min temperature MAE/correlation is 0.071/0.950, compared with 0.233/0.000 for persistence and 0.093/0.909 for linear regression; S3 60 min temperature MAE is 0.190 versus 0.563 and 0.212. Weak cases are also informative: short-window pure illuminance favors persistence (S1 15 min illuminance MAE 5.35 versus 3.42), and SML2010 humidity exposes a measurement-scale mismatch (S2 15 min humidity MAE 0.83 versus 0.20 for persistence).

E. Validity Considerations

The main internal validity risk is that dense full-field results are evaluated in a controlled synthetic room. We mitigate this by reporting IDW, ablations, no-Fourier residual learning, leave-one-scenario-out testing, and a separate real-bedroom sparse-calibration check, but the synthetic layer remains a controlled benchmark rather than measured dense ground truth. The strongest current claim is therefore not that the model has been universally validated in arbitrary homes, but that the proposed decomposition is internally consistent, improves over geometry-free interpolation under controlled full-field truth, and can be calibrated from sparse real observations.

The external validity risks are also explicit. The bedroom data cover one room and one held-out pillow location, so they support point-level calibration plausibility rather than room-wide real 3-D reconstruction. Public datasets give useful external pressure tests but lack the same appliance states, geometry, and dense spatial labels; they are consequently used only for task-aligned comparisons. Finally, recommendation actions are ranked by counterfactual simulation. A deployed closed-loop system must still measure before/after interventions, user comfort, and possible side effects such as energy use or overshooting.

VI. Discussion and Conclusion

The results suggest that appliance-aware reduced-order modeling is more informative than geometry-free interpolation under sparse sensing. The architecture was not selected only for theoretical elegance: early local-only formulations underestimated whole-room AC settling, while IDW ignored source geometry and directionality. The bulk-plus-local split, sensor residual correction, and residual neural layer address these observed failures while keeping the system lightweight.

The evidence should be interpreted at the correct level. Synthetic scenarios provide controlled full-field ground truth and ablations; public datasets provide task-aligned external positioning; the bedroom snapshots show that real sparse observations can improve a held-out pillow reference point. The hybrid residual results also show structured residual learnability within the defined scenario family, not unrestricted generalization to arbitrary buildings. The current evidence does not yet constitute a dense room-scale ground-truth campaign, and action rankings remain counterfactual recommendations until before/after interventions are collected. Future work will focus on longer ESP32 deployment, measured intervention validation, larger real datasets, humidity scale alignment, broader service deployment, and eventually closed-loop control.

Acknowledgment

The authors thank the Department of Computer Science and Information Engineering, National Changhua University of Education, for supporting the thesis prototype development.

References

[1] F. Tao, H. Zhang, A. Liu, and A. Y. C. Nee, “Digital twin in industry: State-of-the-art,” IEEE Transactions on Industrial Informatics, vol. 15, no. 4, pp. 2405–2415, 2019.

[2] A. Fuller, Z. Fan, C. Day, and C. Barlow, “Digital twin: Enabling technologies, challenges and open research,” IEEE Access, vol. 8, pp. 108 952–108 971, 2020.

[3] A. S. Cespedes-Cubides and M. Jradi, “A review of building digital twins to improve energy efficiency in the building operational stage,” Energy Informatics, vol. 7, p. 11, 2024.

[4] P. Bacher and H. Madsen, “Identifying suitable models for the heat dynamics of buildings,” Energy and Buildings, vol. 43, no. 7, pp. 1511–1522, 2011.

[5] P. Hietaharju, M. Ruusunen, and K. Leiviskä, “A dynamic model for indoor temperature prediction in buildings,” Energies, vol. 11, no. 6, p. 1477, 2018.

[6] G. Gokhale, B. Claessens, and C. Develder, “Physics informed neural networks for control oriented thermal modeling of buildings,” Applied Energy, vol. 314, p. 118852, 2022.

[7] E. J. Teshome and F. Haghighat, “Zonal models for indoor air flow - a critical review,” International Journal of Ventilation, vol. 3, no. 2, pp. 119–129, 2004.

[8] A. Megri, Y. Yu, R. Miao, and X. Hu, “A new dynamic zonal model with air-diffuser (doma) - application to thermal comfort prediction,” Indoor and Built Environment, vol. 31, no. 7, pp. 1738–1757, 2022.

[9] B. Huljak, J. A. Acero, Z. H. Kyaw, and F. Chinesta, “Hybrid models for simulating indoor temperature distribution in air-conditioned spaces,” Frontiers in Built Environment, vol. 11, p. 1690062, 2025.

[10] W. Qian, C. Li, H. Gao, L. Zhuang, Y. Lu, S. Hu, and J. Liu, “Estimating indoor air temperature and humidity distributions by data assimilation with finite observations: Validation using an actual residential room,” Building and Environment, vol. 269, p. 112495, 2025.

[11] Y. L. Chen and J. Wen, “Application of zonal model on indoor air sensor network design,” in Sensors and Smart Structures Technologies for Civil, Mechanical, and Aerospace Systems 2007, ser. Proceedings of SPIE, vol. 6529, 2007, p. 652911.

[12] D. Shepard, “A two-dimensional interpolation function for irregularly-spaced data,” in Proceedings of the 1968 ACM National Conference, 1968, pp. 517–524.

[13] Model Context Protocol, “Model context protocol documentation,” https://modelcontextprotocol.io/, accessed: 2026-04-10.

[14] P. Romeu-Guallart and F. Zamora-Martinez, “SML2010 [dataset],” UCI Machine Learning Repository, 2014.

[15] G. Chitalia and M. Pipattanasomporn, “CU-BEMS, smart building electricity consumption and indoor environmental sensor datasets,” Scientific Data, vol. 7, p. 290, 2020.